



Parktown Boys' High School

Geography Department

Grade 12 Control Test

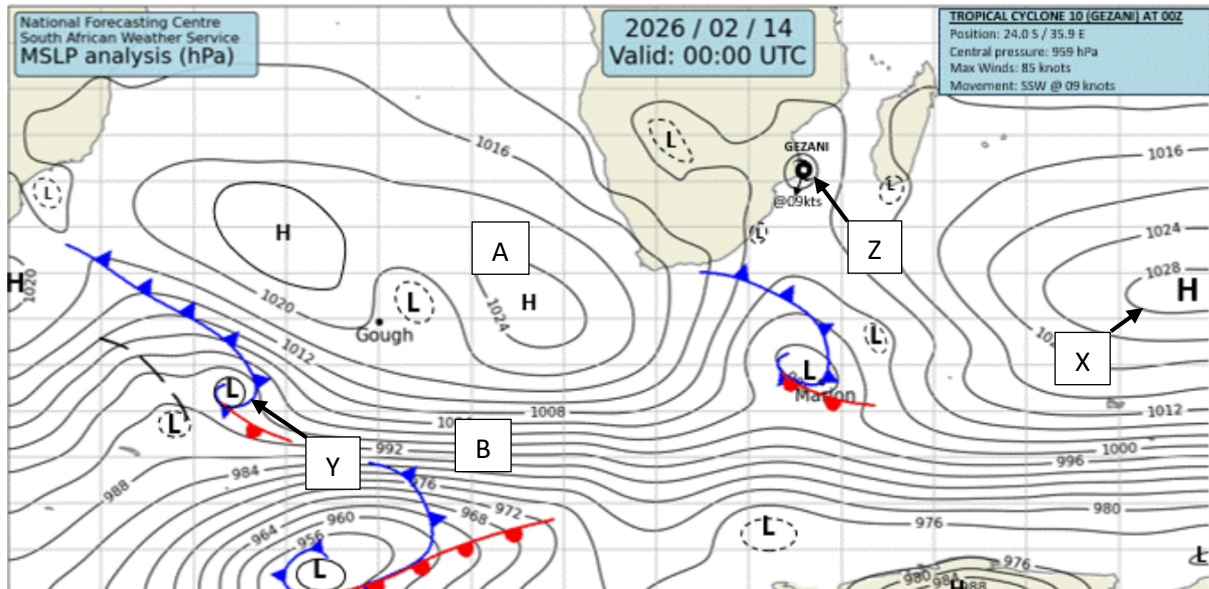
MEMO

INSTRUCTIONS AND INFORMATION

1. The question paper consists of FOUR questions.
2. ALL diagrams are included in the question paper.
3. Where possible, illustrate your answers with labelled diagrams.
4. Leave a line between subsections answered.
5. Number your answers correctly according to the numbering system used in this question paper.
6. Write neatly and legibly.

Question 1 – Short Questions [15]

1.1. Refer to the synoptic chart below. Answer the questions that follow by choosing only the correct term in brackets. E.g. 1.9. Updog



- 1.1.1. The season shown in the synoptic chart is (Summer/Winter/Autumn/Spring).
- 1.1.2. The high pressure at X is the (South Atlantic / South Indian / Kalahari) high pressure.
- 1.1.3. (Z/Y) is a mid-latitude cyclone.
- 1.1.4. (Z/Y) is a tropical cyclone.
- 1.1.5. The wind speed at B is fast. We know this because (it has a steep pressure gradient / Coriolis force is powerful here / as pressure increase so does the wind speed).
- 1.1.6. The conditions at A are likely (clear and calm / wet and windy / cold and humid).
- 1.1.7. Tropical storm Gezani is the (2nd / 5th / 7th) cyclone of the season.
- 1.1.8. Tropical storm Gezani, according to the chart, is currently over (Mozambique / Madagascar / South Africa).

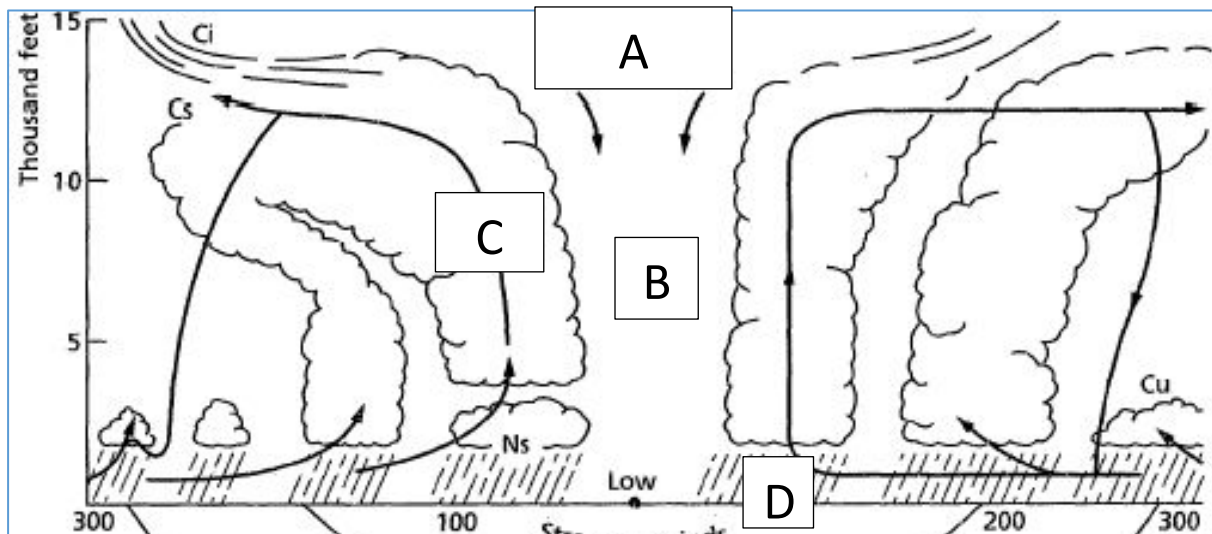
(8x1) (8)

Question 1.1

1	Summer
2	South Indian

3	Y
4	Z
5	it has a steep pressure gradient
6	Clear and calm
7	7th
8	Mozambique

1.2. Refer to the figure below and statements that follow. Select only the correct word in brackets in each statement to make the statement TRUE.



- 1.2.1. The air at A is (diverging / subsiding / low pressure) air.
- 1.2.2. When B is fully formed and visible on satellite imagery, we would say the cyclone is in its (formative / mature / decaying) stage.
- 1.2.3. The weather experienced at B is (violent with heavy rains / clear and calm / extremely windy, but with little to no rain).
- 1.2.4. The clouds that typically form at C are (cumulonimbus / stratus / cirrus) clouds.
- 1.2.5. These systems occur between latitudes of (0° - 15°S / 5° - 30°S / 30° - 60°S) in the southern hemisphere.
- 1.2.6. The storm surge at D is caused by (tsunamis / very low pressure / heavy rain).
- 1.2.7. Cape Town (always / occasionally / never) experiences hurricane strength winds from these storms.

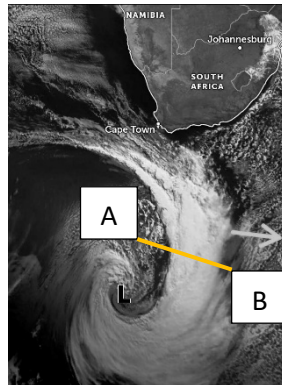
(7x1) (7)

Question 1.2

1	Subsiding
2	Mature
3	Clear and calm
4	Cumulonimbus
5	5° - 30°S
6	Very low pressure
7	Never

Question 2 – Mid-latitude Cyclones [15]

Refer to the satellite image below and answer the questions that follow.



2.1. What phenomenon is being shown in the image? (1x1) (1)

Mid-latitude cyclone ✓

2.2. What are the names of the two wind masses that meet at the Polar Front to form mid-latitude cyclones? (2x1) (2)

Westerlies ✓ and Polar Easterlies ✓

2.3. Why are mid-latitude cyclones more common in winter along the southern coast of South Africa? (1x2) (2)

[Accept any ONE]

Northern hemisphere is receiving more heat and so ITCZ migrates north ✓✓

South Atlantic and South Indian HPs migrate north ✓✓

Polar front migrates north ✓✓

Westerly wind belt shifts north ✓✓

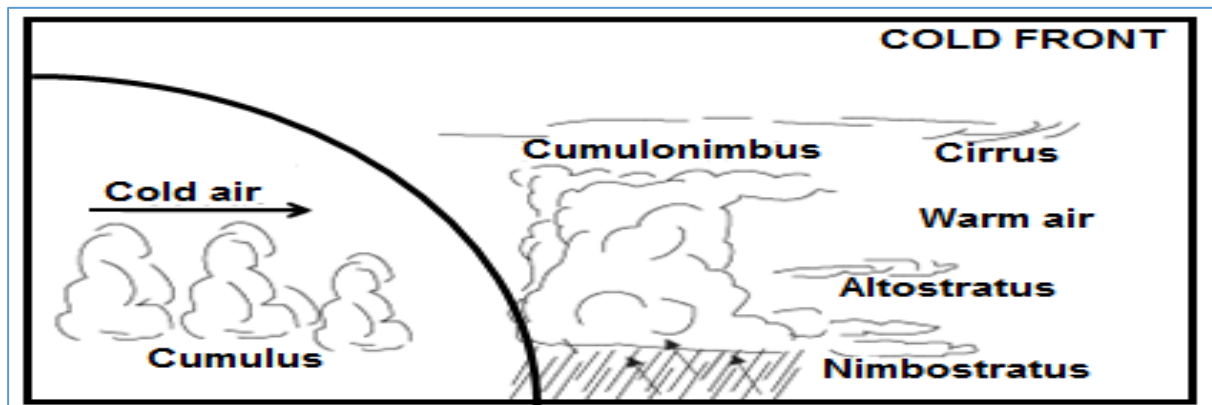
2.4. Draw a fully labelled cross section from A to B. Make sure you have the following:

Front shape

Temperature of air pockets

Cloud types at front

(4x1) (4)



Front shape ✓

Temperature of air pockets (cold ✓ and warm ✓)

Cloud types at front ✓

2.5. Describe how the temperature and pressure in Cape Town is going to change as the mid-latitude cyclone passes over.

(3x2) (6)

[Any THREE acceptable answers – MUST MENTION TEMP AND PRESSURE]

Cape Town experiences warmer conditions initially. ✓✓

The temperature falls as the cold air approaches. ✓✓

The pressure will drop (accept air rises) ✓✓

Pressure increases when the cold front passes over (accept cold air sinks over Cape Town) ✓✓

Question 3 – Tropical Cyclones [15]

Refer to the write-up of Tropical Cyclone Gezani below and answer the questions that follow.

The SAWS Monitoring Moderate Tropical Storm Gezani in the South-West Indian Ocean

The South African Weather Service (SAWS) is closely monitoring a moderate tropical storm named Gezani, currently located over the south-west Indian Ocean to the east of Madagascar. At present, there is no indication of any direct threat to South Africa.

Gezani is currently moving westward under favourable environmental conditions, and forecast guidance indicates that landfall over eastern Madagascar is likely on Tuesday afternoon, 10 February 2026. As the system approaches Madagascar, environmental conditions may support further intensification, and Gezani could strengthen into a tropical cyclone, or possibly an intense tropical cyclone, before landfall.

Following landfall, the system is expected to weaken while tracking south-westward across Madagascar. Gezani is then forecast to enter the Mozambique Channel late on Wednesday, 11 February 2026, where environmental conditions may once again become favourable for re-intensification. Some forecast scenarios suggest a possible approach toward the Mozambique coast later in the week; however, uncertainty remains regarding the exact track and intensity during this phase.

The SAWS advises vessels operating in or transiting through the south-west Indian Ocean and the Mozambique Channel, as well as offshore marine operations to exercise caution, as rough seas, strong winds, and hazardous marine conditions are expected in the vicinity of the storm.

- 3.1. Which ocean did Cyclone Gezani pass through while moving towards Mozambique? (1x1) (1)

Indian Ocean ✓

- 3.2. According to the article, why should vessels operating in the south-west Indian Ocean and Mozambique Channel exercise caution? (2x1) (2)

[Any TWO]

Rough seas, strong winds, and hazardous marine conditions are expected in the vicinity of the storm. ✓✓

- 3.3. Gezani is expected to weaken as it passes over Madagascar.
Provide TWO reasons as to why this will happen. (2x2) (4)

[Accept reasonable answers]

Friction increases over land decreasing wind speed ✓✓

Dry air enters the system ✓✓

No longer over warm ocean ✓✓

Air pressure increases ✓✓

- 3.4. Discuss TWO economic impacts that Gezani would have on Madagascar, and provide TWO solutions that Madagascar could implement to reduce the impact of future cyclones. (4x2) (8)

[Any TWO – Must be ECONOMIC impacts! Can accept other reasonable answers]

Economic Impacts:

Destruction of Infrastructure: Roads, bridges, and communication networks may be damaged, disrupting trade and daily economic activities. ✓✓

Agricultural Losses: Heavy rains and strong winds can destroy crops and kill livestock, leading to food shortages and loss of income for farmers. ✓✓

Business Disruptions: Flooding and storm damage can force businesses to close, resulting in job losses and reduced economic productivity. ✓✓

Tourism Decline: Damage to coastal resorts and transport infrastructure can discourage tourists, leading to revenue losses in the tourism sector. ✓✓

[Any TWO – Can accept other reasonable answers]

Solutions:

Prediction and Warning: Invest in systems that can communicate effectively to the people allowing for smooth evacuation. ✓✓

Improved Disaster Preparedness (Preparing the community): Strengthening early warning systems and evacuation plans can help reduce losses. ✓✓

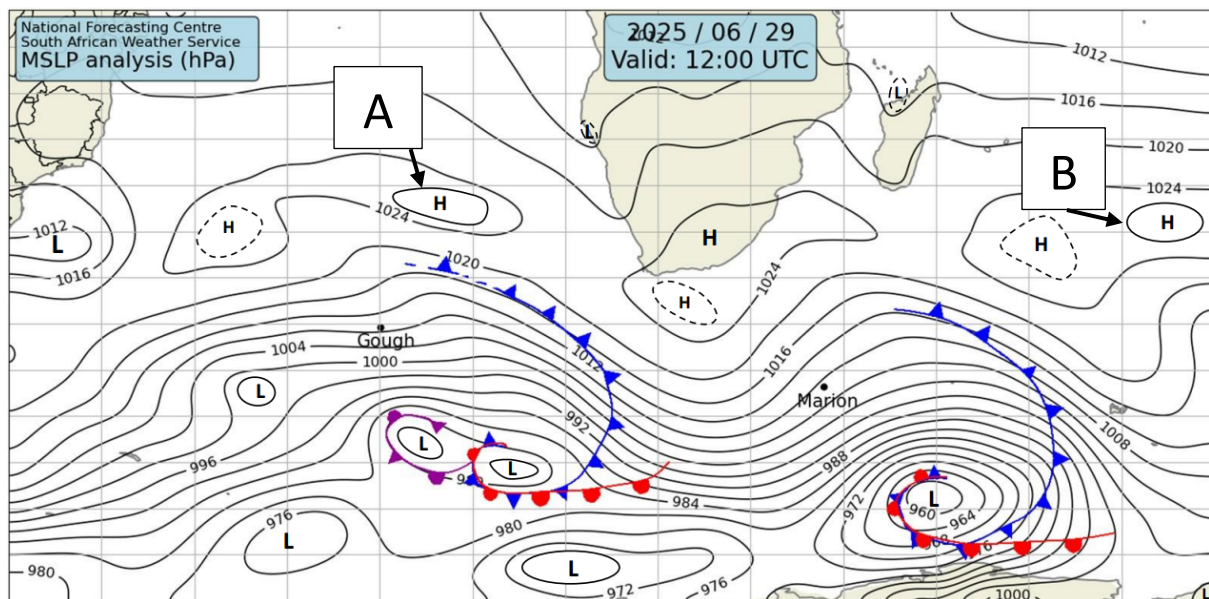
Stronger Infrastructure: Constructing cyclone-resistant buildings, roads, and flood barriers can minimize damage during future storms. ✓✓

Reforestation and Coastal Protection: Planting mangroves and preserving natural barriers can help reduce the impact of storm surges and flooding. ✓✓

Financial Support for Affected Communities: Establishing emergency funds and insurance schemes can help businesses and farmers recover more quickly. ✓✓

Question 4 – Anticyclones [15]

Refer to the synoptic chart below and answer the questions that follow.



- 4.1. What season is depicted in the synoptic chart? Provide TWO pieces of evidence to substantiate your answer. (1+2) (3)

Winter ✓

[Accept any TWO]

Date, presence of Kalahari HP, presence of mid-lat cyclones close to SA. ✓✓

- 4.2. List TWO characteristics of an anticyclone. (2x1) (2)

Dry, descending air ✓

Clear, cloudless skies ✓

- 4.3. Name the anticyclone present over South Africa. (1x1) (1)

Kalahari ✓

- 4.4. Name the TWO anticyclones visible at A and B on the chart. (2x1) (2)

South Atlantic ✓ and South Indian anticyclones ✓ [NOTE: MUST include "South"]

- 4.5. What is the actual pressure in hectopascals (or millibar) in at A? (1x1) (1)

1028 hPa ✓

4.6. Explain why all of these anticyclones are situated here at this time. (1x2) (2)

[Accept reasonable answers]

Northern hemisphere is receiving more direct sun (or visa versa) ✓✓

Cooler conditions experienced by Southern Hemisphere at this time ✓✓

ITCZ, and all pressure belts shifted north. ✓✓

4.7. Explain how the position of the anticyclones over South Africa and mid-latitude cyclones depicted in the chart lead to typical weather conditions across South Africa during this season. (2x2) (4)

[Any TWO reasonable responses]

Anti-cyclones

Leads to clear skies across interior ✓✓

Prevents rising air ✓✓

Prevents moisture entering interior ✓✓

Mid-latitude cyclones

Leads to winter rainfall across the Western Cape ✓✓

Brings cold conditions to SA ✓✓

GRAND TOTAL: 60 MARKS